

Question Paper

Exam Date & Time: 12-Nov-2019 (10:00 AM - 01:00 PM)



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

1st Semester of M.Sc.(Physics) Examination November-2019

Date: 12/11/2019(Tuesday) Time: 10.00 a.m.-1.00 p.m.

MATHEMATICAL METHODS OF PHYSICS [PS725]

Marks: 70

Duration: 180 mins.

Multiple Choice Questions

Answer all the questions.

Section Duration: 30 mins

Q-I Choose the correct answer the following questions.

- 1) A function $f(z) = |z|^2$ is _____. (1)

[analytic only at origin](#) [analytic everywhere in finite complex plane](#) [nowhere analytic](#) [not continuous at origin](#)

- 2) A function $u(x, y)$ is said to be harmonic if _____. (1)

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial y^2} \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0 \quad \frac{\partial u}{\partial x} - \frac{\partial u}{\partial y} = 0$$

- 3) A mapping $w = \frac{1}{z}$ is called _____. (1)

[magnification](#) [bilinear](#) [inversion](#) [rotation](#)

- 4) If $f(z)$ is _____, then a mapping $w = f(z)$ is conformable. (1)

[analytic](#) [continuous](#) [differentiable](#) [none of these](#)

- 5) What is the value of integral $\int_C \frac{z^2 - z + 1}{z - 1} dz$, where C is the circle $|z| = \frac{1}{2}$? (1)

[π](#) [2πi](#) [0](#) [1](#)

- 6) A function $\frac{z^2 - 3}{(z - 3)^2}$ has _____ at $z = 3$ (1)

[essential singularity](#) [non-isolated singularity](#) [pole of order 2](#) [simple pole](#)

- 7) The residue of the function $\frac{1 - 2z}{z(z - 1)(z - 2)}$ at $z = 1$ is _____. (1)

[1/2](#) [-3/2](#) [2](#) [1](#)

- 8) The region of validity of $\frac{1}{1+z}$ for its Taylor's series expansion about $z = 0$ is _____. (1)

[|z| < 1](#) [|z| > 1](#) [|z| = 1](#) [none of these](#)

- 9) A function which is analytic everywhere in a complex plane is known as _____. (1)

[a regular function](#) [an entire function](#) [a harmonic function](#) [none of these](#)

- 10) If a square matrix A of order n has linearly independent eigen vectors, then a matrix P can be found such that _____ is a diagonal matrix. (1)

$$AP^{-1} \quad P^{-1}AP \quad A^{-1}PA \quad PA^{-1}$$

- 11) The eigenvalues of a 4×4 matrix A are 2, 3, 13 and 7. Then determinant of A is _____. (1)

$$1 \quad 25 \quad 15 \quad 546$$

- 12) The Euclidean norm of a vector (5,-1,1) is _____. (1)

$$\sqrt{3} \quad 3\sqrt{3} \quad 5 \quad 1$$

- 13) Let V be an inner product space with inner product $\langle \cdot, \cdot \rangle$ and let $x, y \in V$. Then x and y are said to be orthogonal if _____. (1)

$$\langle x, y \rangle = 0 \quad \langle x, y \rangle \neq 0 \quad \langle x, y \rangle = 1 \quad \langle x, y \rangle \neq 1.$$

- 14) The dimension of a real vector space $R_n[x] = \{a + bx + cx^2/a, b, c \in R\}$ is _____. (1)

$$1 \quad 2 \quad 3 \quad 4$$

- 15) Which of the following is the kernel of Fourier sine transform? (1)

$$\sqrt{\frac{2}{\pi}} \cos sx \quad \sin sx \quad \sqrt{\frac{2}{\pi}} \sin sx \quad \cos sx$$

- 16) If $F[f]$, $F_s[f]$ and $F_c[f]$ are Fourier transform, Fourier sine transform and Fourier cosine transform of an even integrable function f respectively, then _____. (1)

$$F[f] = F_c[f] \quad F[f] = F_s[f] \quad F[f] = -iF_s[f] \quad F[f] = iF_c[f]$$

- 17) The inverse of $-i$ in the multiplicative group $\{1, -1, i, -i\}$ is _____. (1)

$$1 \quad i \quad -i \quad 1$$

- 18) Which of the following is a group with usual multiplication operation? (1)

$$\{R, \times\} \quad \{N, \times\} \quad \{Z, \times\} \quad \text{none of these}$$

- 19) Which of the following is covariant tensor of order two? (1)

$$A_j^i \quad A^{ij} \quad A_{ij} \quad \text{none of these}$$

- 20) Which of the following is Christoffel symbol of the first kind? (1)

$$[ij, k] \quad \left\{ \begin{matrix} k \\ ij \end{matrix} \right\} \quad g_{ij} \quad ds^2$$

SECTION-I

Answer all the questions.

Q-II Answer the following questions as directed.

- 21) Find the inverse of a matrix $\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$ if exist. (3)

- 22) Show that vectors (2,-1,3), (4,1,-1) and (2,3,-3) are linearly independent. (2)

- [OR]
23) Show that a linear transformation $T: \mathbb{C} \rightarrow \mathbb{R}^2$ define by $T(a + ib) = (a, b)$ is isomorphism. (2)

- 24) (2)

Using Cauchy's integral formula, evaluate the integral $\int_C \frac{2z+1}{z^2+z} dz$, where C is $|z| = \frac{1}{2}$.

25) Discuss the analyticity of $f(z) = \bar{z}$, where $\bar{z}$ is the conjugate of z . (2)

[OR]
26) Find the residue of a function $\frac{12z-7}{(z-1)^2(2z+3)}$ at $z = 1$. (2)

27) Find the Fourier transform of $f(x) = e^{-a|x|}$, $a > 0$. (3)

[OR]
28) Find the Fourier cosine transform of $f(x) = \begin{cases} 25 - x^2, & -5 \leq x \leq 5 \\ 0, & \text{otherwise.} \end{cases}$ (3)

29) Write the terms contained in $S = a_{ij}x^i x^j$ taking $n = 4$. (3)

[OR]
30) Explain contravariant tensor and covariant tensor with example. (3)

31) If x^i and $\bar{x}^i$ are independent coordinates of a point, then show that (2)

$$\frac{\partial x^j}{\partial \bar{x}^k} \cdot \frac{\partial \bar{x}^k}{\partial x^i} = \delta_i^j$$

[OR]
32) Define: outer product of two tensors. (2)

33) Show that $\{1, -1, i, -i\}$ forms a group with multiplication. (3)

[OR]
34) Explain group isomorphism with an example. (3)

SECTION-II

Answer all the questions.

Q-III Answer the following questions as directed

35) Show that R^n is a vector space over R . (5)

36) Find the eigen values and eigen vectors of the matrix $\begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$ (5)

[OR]
37) Show that a transformation $T: R^2 \rightarrow R^2$ defined by (5)

$T(x, y) = (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta)$ is linear. Also find matrix of T with respect to bases $\{(3, 0), (0, -2)\}$ and $\{(1, 0), (0, 1)\}$.

38) Show that the function $u(x, y) = x^4 - 6x^2y^2 + y^4$ is harmonic. Also find the analytic function $f(z) = u(x, y) + iv(x, y)$. (5)

[OR]
39) Expand $f(z) = \frac{1}{(z+2)(z+5)}$ in Laurent series valid for (i) $|z| < 2$, (5)

(ii) $2 < |z| < 5$ and (iii) $|z| > 5$

40) Using Cauchy residue theorem, evaluate the integral $\int_0^{2\pi} \frac{d\theta}{5-3 \cos \theta}$ (5)

- [OR]
41) Find the bilinear transformation which maps the points $0, -1, i$ in z -plane onto $i, 0, \infty$ in w -plane. (5)
Also find the image of the unit circle $|z| = 1$.
- 42) Define: Fourier transform. Find the Fourier transform of a Gaussian function (5)
 $f(x) = e^{-kx^2}, k > 0. (\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi})$
- [OR]
43) Solve $\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2} (x > 0, t > 0)$ subject to the conditions $u(x, 0) = 0$ and $\frac{\partial u}{\partial x}(0, t) = -a$ (5)
, where a is constant and $u, \frac{\partial u}{\partial x} \rightarrow 0$ as $x \rightarrow \infty$.
- 44) Discuss the group of symmetries of an equilateral triangle. (5)
- [OR]
45) Discuss the group representation of a group of symmetries of a square. (5)

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